

PRATEEK RAI SRIVASTAV

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SUMMARY

Experienced Software Developer with a background in contributing to CNCF Landscape open-source projects and enhancing software performance using **AWS services, GitOps, and MLOps**. Previously **led a team at Apna Beta** to design **scalable** infrastructure on AWS, improving system capacity to handle a **1 Million** requests per day. Successfully integrated Kafka for data streaming at ADP, boosting system performance by **30%**. Eager to leverage expertise in software development and infrastructure optimization to drive innovation as a Software Engineer.

TECHNICAL SKILLS

Languages: Java, Python, Go, HTML/CSS, JavaScript, SQL, NoSQL, Linux Shell Scripting (Bash)

Frameworks: React JS, Spring Framework, Spring Boot, Spring JPA, Hibernate ORM, Spring Security, Angular

Achievements: **SWE Fellow at HeadStarter AI, Volunteer at Google Developer Student Clubs**

Deployment Tools: Git, AWS, Docker, Kubernetes, Terraform, Jenkins CI/CD, GitHub Actions, Argo CD, Grafana

PROFESSIONAL EXPERIENCE

University Of Michigan, On-Site, United States: Graduate Teaching Assistant Jan 2025 – Present

- Educated and guided **60 students on Docker, Kubernetes, and AWS EKS**, incorporating practical labs to build and manage k3s clusters.
- Fostered student engagement by mentoring on open-source contributions, emphasizing collaboration, version control practices, and industry-relevant skills.

University Of Michigan, On-Site, United States: Graduate Research Assistant Oct 2024 – Present

- Built a Machine Learning Model to monitor CPU usage of home routers using **Recurrent Neural Networks (RNN)** with **numpy, Tensorflow and matplotlib**, which improved task offloading to the network edge. Applied **scheduling and concurrency algorithms** to offload multiple tasks, optimizing router usage and enhancing network efficiency using **Web Assembly**.

AWS OpenSearch (Contributor), Remote, United States: Software Developer Oct 2024 – Present

- Configured rebasing strategy for Dependabot PRs in Python and developed new Jenkins pipeline using **Groovy** reducing manual intervention in Git and dependency on maintainers, thereby actively contributing to the OpenSearch project under the CNCF landscape

ADP, Hyderabad, India: Software Developer I Feb 2022 – Sept 2023

- **WGRS-Wage Garnishments Record System** Part of the core development team. **Java** and **SpringBoot, Maven, Oracle SQL** to build robust backend services. Developed **RESTful APIs** and applied multiple **design patterns** and utilized the **microservices** software architecture. Integrated **Kafka** for reliable **message queuing and data streaming**, along with **Batch Processing** using **Spring Batch** enhancing the system's scalability and performance by **30%**
- Simplified the onboarding process, reducing the time required to less than two weeks, which led to a **150% increase in productivity**. Followed **Agile methodologies**, principles and documented the product in **Confluence**

Apna Beta, Hyderabad, India: Software Development Lead Sep 2020 – Feb 2022

- As a Lead Architect in a 5-member startup, I designed and deployed scalable AWS infrastructure using **Terraform**, reducing cloud costs by **30%** while handling up to 1 million requests per day. I built **CI/CD pipelines with Jenkins**, streamlining deployments of **containerized microservices on AWS EKS** and cutting deployment time by **50%**. Implementing best practices like **TDD, BDD, and code reviews** improved code quality and maintainability. I enhanced system monitoring with **Grafana, CloudWatch, and Splunk**, reducing incident resolution time by **40%**.

EDUCATION

University Of Michigan, Dearborn: Master's In Computer and Information Science Aug 2023 – April 2025